

The Changing Latitude: Labour-Sponsored Venture Capital Corporations in Canada

Sofia Johan

York University - Schulich School of Business
4700 Keele Street, Toronto, Ontario, Canada
<http://ssrn.com/author=370203>
sjohan@schulich.yorku.ca

and

Tilburg Law and Economics Center (TILEC)
Warandelaan 2, Tilburg, 5000 LE, Netherlands

Denis Schweizer

WHU - Otto Beisheim School of Management
56179 Vallendar, Germany
<http://ssrn.com/author=744422>
Denis.Schweizer@whu.edu

Feng Zhan

York University - Schulich School of Business
4700 Keele Street, Toronto, Ontario, Canada
<http://ssrn.com/author=1570631>
fzhan@schulich.yorku.ca

JEL Classification: G3, G38, H3

Keywords: Venture capital, LSVCC, corporate governance, tax policy, investments

* We owe thanks to Rajesh Chakrabarti, Douglas Cumming, Millan Jankovic and the seminar participants of the CGIR Special Issue Conference on "Global Perspectives on Entrepreneurship: Public and Corporate Governance" at the Schulich School of Business, York University, Toronto, April 25-26, 2013.

The Changing Latitude: Labour-Sponsored Venture Capital Corporations in Canada

ABSTRACT

Manuscript Type: Empirical

Research Question/Issue: This paper seeks to understand the role corporate governance and government policy plays in the portfolio choices of the Labour-Sponsored Venture Capital Corporations (LSVCCs) in Canada. We investigate whether or not the change in tax policy announced in Ontario (2005) had an impact on the investment behaviour of Ontario LSVCCs and whether the unique corporate governance structure of LSVCCs enables them to focus on their investment mandate subsequent to this announcement.

Research Findings/Insights: Our finding suggests that LSVCCs in Ontario are more likely to include public companies in their fund portfolios after the announcement of the change in tax policy. We find that after 2005, LSVCCs have increased their number of investments in public companies by 59.13% and in turn decreased their number of investments in private companies by 13.17%. On the other hand, we find no significant changes of in investment behaviour for LSVCCs in other provinces. In terms of the percentage of total investment in public companies, we find that the LSVCCs in Ontario are more likely to increase their total investment in public companies by 50% and to decrease their investment in the short term by 46.43%. LSVCCs in other provinces however are reducing their percentage of investment in public companies by 58.33% and increasing their total investment in private entrepreneurial firms by 38.33% in the same period.

Theoretical/Academic Implications:

With a hand-collected proprietary dataset, we are able to augment existing studies on the unique structure of LSVCCs in Canada with empirical evidence on the style drift due to the changes in government tax policy. We compare and contrast the investment behaviour of LSVCCs before and after the tax policy change in Ontario as well as the investment behaviour of LSVCCs in other provinces. We hypothesize that as a result of the elimination of the tax credits, the removal of certain investment restrictions, and weaker corporate governance LSVCCs have drifted from their original mandate to invest in high-risk venture companies to investing in less risky public companies. Such style drift may be a result of LSVCCs preparing for potential wealth transfer or liquidation by retail investors. More importantly, we find the unique corporate governance structure of LSVCCs may facilitate this drift from their original purpose of providing venture capital to small and medium-sized entrepreneurial (SME) firms.

Practitioner/Policy Implications:

We highlight that the style drift of LSVCCs in Ontario may result in such funds behaving more like other types of mutual funds and the deviation from their original mandate to provide venture capital may not only prove detrimental to entrepreneurial investee firms seeking such capital, but also negate any diversification benefits sought by fund investors. Also, such deviation may not necessarily justify the higher management expense ratio charged by LSVCCs.

INTRODUCTION

The economic benefits of venture capital have been well established by numerous academic studies. Research has established that in addition to bridging the capital gap in the financing of entrepreneurial firms, venture capital fund managers play a significant role in enhancing the value of their entrepreneurial investments as they provide considerable administrative, marketing, and strategic advice to entrepreneurial firms, as well as facilitating a network of support for an entrepreneurial firm with access to accountants, lawyers, and investment bankers (Cumming and Johan, 2009; Sahlman, 1990; Sapienza *et al.*, 1996; Gompers and Lerner, 1999; Manigart *et al.*, 2002a,b; Leleux and Surlemont, 2003; Wright and Lockett, 2003). To encourage venture capital investment in Canada, Canadian Labour-Sponsored Venture Capital Corporations (LSVCCs, also known as Labour Sponsored Investment Funds (LSIFs)) were created with the objective of providing venture capital to small and medium-sized entrepreneurial (SME) firms to facilitate economic development and growth. Unlike other more traditional venture capital funds, the LSVCCs were created and structured to raise capital from the large retail investor pool and not institutional investors, and too facilitate this objective, both the federal and provincial governments provide tax credits for individual investors who invest in these funds. The tax credits are also provided to incentivize investments in such funds as while they are seen to be potentially particularly beneficial to the economy in terms of increasing innovation and other benefits, they are also deemed to be more risky than mutual funds. It is interesting, however, that several studies (Sandler, 2004; Cumming and MacIntosh, 2006; Cumming, 2007; Cumming and MacIntosh, 2007) have found that not only do LSVCCs provide retail investors with lower returns but the performance of LSVCCs are also far behind many

other indices, such as small cap fund indices as well as the lowest risk Treasury bills. These findings cast doubt on the success of the government sponsored program and question the ability of these LSVCCs to provide long-term economic growth. More importantly, research suggests that LSVCCs have a negative impact on the overall capital available to the entrepreneurial firms as the tax credits awarded and the fund management mandates or constraints structured as such that the access to cheaper capital and the necessity for such capital to be invested within a tight time frame may have resulted in a "crowding out" of other types of venture capital investors (Cumming and Johan, 2009; Cumming and MacIntosh, 2006). This not only directly undermines the initial purpose of creating the LSVCCs and may also exacerbate direct losses in government tax revenue.

As a result, the provincial government of Ontario announced on September 30, 2005 that it would gradually eliminate the tax credit for individual investors investing in LSVCCs. It also lifted the restriction on the investment in publicly listed companies by LSVCCs. Eliminating the tax credit would undoubtedly reduce the attractiveness of the venture capital funds to retail investors therefore impacting the inflow of fund capital into LSVCCs in Ontario. In this paper, we seek to understand the roles corporate governance structure and tax policy in Ontario played in the portfolio choices of the LSVCCs. We investigate whether or not the announcement of the elimination of the tax credit in Ontario changed the investment behaviour of Ontario LSVCCs and whether the unique corporate governance structure encourages LSVCCs to maintain their original function of providing venture capital to private entrepreneurial firms.

The elimination of tax credits to investors in Ontario will have an impact on LSVCC fund inflow. Studies have shown that investors are incentivized to invest in LSVCCs with the promise of tax credits due to the cheaper actual cost of initial investment. More importantly, the same

investors are less concerned about the long term after tax return due to the up-front benefits (Cumming and MacIntosh, 2003 a, b; Cumming and Johan, 2009; Barclay et al., 1998; Bergstresser and Poterba, 2002). LSVCCs in Ontario therefore will become less attractive to retail investors after the change in tax policy and there will in turn be added pressure for higher returns from existing investors as they get closer to their allowed redemption dates or when their statutory lock-up periods expire. LSVCCs have been found to be underperforming even risk free 30 day Treasury Bills (Cumming and MacIntosh, 2006, 2007; Cumming and Johan, 2009), and as a result fund managers in Ontario are incentivised to alter their investment behaviour to counter or mitigate the potential effect of the change in investor sentiment pursuant to the removal of the tax credit, and the potential redemptions as the lock-up periods expire. The fund managers may alter portfolio risk, measured by volatility, by drifting away from the less volatile investments in private entrepreneurial firms towards more volatile public companies, or the fund manager could continue to provide venture capital to risky firms with huge potential upside returns (positively skewed returns distribution) as their initial investment mandate required. Recall that LCVCC fund managers have distinct skill sets for providing venture capital and value added to entrepreneurial firms and therefore they may not be as skilled in investing in publicly listed companies as mutual fund managers or hedge fund managers. Also, in terms of market risk, volatility is higher for public equity compared to private investments (see Cumming et al., 2013).

Our findings suggest that pursuant to the announcement, LSVCCs in Ontario are increasingly reallocating their assets from private firms toward public companies. We find that after 2005, LSVCCs increased the number of public companies in their portfolios by 59.13% and decreased (and/or omissions of reinvestments from exited private companies) the number of private companies by 13.17%. These results are statistically significant at the 10% and 1%

levels, respectively. On the other hand, we find no evidence to suggest the same shift in terms of the fund portfolios for LSVCCs in other provinces within the same period. In terms of percentage of total investment in public companies, we find that while the LSVCCs in Ontario are more likely to increase their investments in public companies by 50%, the LSVCCs in other provinces are decreasing their holdings in public companies by 58.33% since 2005, when compared with the overall average. Our results also indicate that LSVCCs in provinces other than Ontario are likely to increase their total investment in private entrepreneurial firms by 38.33% while LSVCCs in Ontario are more likely to decrease their total investment in the short term by 46.43% compared with the overall level of average investment in short term.

Venture capital funds are known to style drift, for example from the riskier early stage to the latter stages of investment such as mezzanine and buy-out stages, and they are also known to invest in public versus privately held firms (Cumming, Fleming and Schwienbacher, 2009; Chaplinsky and Haushalter, 2012; and Cumming and Johan, 2012). Our research augments such academic work as our results indicate that after the changes in tax policy in 2005, there is a trend for LSVCCs in Ontario to drift away from investing in private entrepreneurial firms and increasingly investing in public companies, both in terms of the number of public companies in the fund portfolio and the percentage of total investment in the fund portfolio. LSVCCs in Ontario may be style drifting away from providing venture capital and moving toward the more liquid investment behavior of other types of mutual funds more commonly utilized by retail investors as investment vehicles. Considering LSVCCs were created with the intention of facilitating the provision of venture capital to entrepreneurial firms, our results suggest that LSVCCs may not be fulfilling their intended function. LSVCCs were also created to enable retail investment in this alternative investment vehicle that is not ordinarily availed to investors other

than institutional and high net worth investors. This alternative investment also enabled retail investors to obtain portfolio diversification benefits, and as such incentives in the form of tax credits were provided by both the federal and provincial governments to make such investments more palatable. Our results also suggest that style drifting by Ontario LSVCCs managers may be misleading retail investors as they would have initially acquiesced to the higher management expense ratios (as compared to mutual funds) because they had invested in a venture capital fund, not a fund similar in strategy to other mutual funds. In addition to not fulfilling their function, they may also not be executing their investment mandates as agreed upon with their investors. Furthermore, we find that the Return on Assets of LSVCCs in Ontario have decreased statistically significantly after the changes in tax policy were announced, while the volatility simultaneously increased. Both results fit into the view that the style drift may be increasing overall portfolio risk of both LSVCC and investor, and not reducing it.

More importantly, our study provides new evidence of the effect of weak corporate governance on LSVCCs. Retail investors are unlike other institutional investors in that their disparate make-up does not allow for efficient oversight of fund managers. That in addition to the disincentives for oversight or demands for excellent returns as a result of tax credits may enable or even encourage fund managers to style drift, take on additional portfolio risk and move away from their original function of providing venture capital.

Our paper contributes to extant literature in the following manner. First, we are the first to examine the effects of a change in government policy on a retail-based government sponsored venture capital program. Our study does not only analyze the effects of tax policy changes on LSVCCs in one jurisdiction, Ontario, but also compares the LSVCCs in Ontario to others across multiple jurisdictions or rather to those situated in and governed by other provincial governments.

Second, we provide empirical evidence that the announcement of the removal of tax credits in Ontario have had an impact on the investment behaviour of LSVCCs in Ontario. Pursuant to the announcement, LSVCCs in Ontario are increasingly shifting away from venture capital investments in private entrepreneurial firms and investing more in publicly listed companies. Third, we do not find similar drifts among LSVCCs across other jurisdictions or provinces and therefore we believe our results have several policy implications. The central purpose of creating LSVCCs was to bridge the capital gap faced by entrepreneurial firms and facilitate the provision of both financial capital and also the more significant value added provided by venture capital fund managers. In addition, the creation of LSVCCs enabled retail investors to invest an alternative investment asset class (venture capital) that would ordinarily only be availed to institutional investors and high net worth individuals, and as a result potentially diversify from their portfolio of mutual funds, stocks and bonds. If LSVCCs are style drifting away from venture capital and increasingly shifting their investments toward public companies without providing superior returns, as supported by our findings, this raises the question of whether or not LSVCCs are fulfilling their economic function, executing their mandates as agreed upon with their investors and generally doing what they are paid to do.

The rest of the paper is organized in the following manner. Section 2 provides a brief description of characteristics of LSVCCs in Canada and provides an overview of the general tax credit provided by both federal and provincial governments to individual retail investors. Section 3 discusses some theoretical arguments that affect fund manager investment behaviour due to the change of government tax policy and introduces our three hypotheses. The data and summary statistics are introduced in Section 4. Section 5 presents the univariate comparison and Section 6 provides the multivariate analyses of the impact on the announcement of the change in the tax

policy in Ontario. Section 7 provides some robustness check and concluding remarks follow in the last section.

LSVCCS IN CANADA AND OVERALL TAX CREDITS TO LSVCC INVESTORS

The development and characteristics of LSVCCs, the primary government support mechanism for the provision of venture capital in Canada since the 1980's, have been thoroughly researched in several studies (Cumming and MacIntosh, 2002, 2003, and 2006 and Cumming and Johan, 2009). Here, we briefly summarize the unique characteristics of LSVCCs that were created and structured with the principal purpose of increasing the pool of capital to finance small entrepreneurial firms and to promote local economic development. The first LSVCC was formed in the province of Quebec in 1983. Other provinces in Canada were quick to follow with their own legislation to form their own LSVCCs.

While the LSVCCs across provinces may not be identical in both legal and operational structure, there are some common characteristics among all LSVCCs across Canada. First, unlike other venture capital funds that are usually formed as limited liability partnerships, LSVCCs are formed as corporations. Second, similar to mutual funds and not like other forms of alternative investments, investors in LSVCCs are individual retail investors, and not sophisticated investors such as institutional investors or high net worth individuals. Unlike mutual funds where there is a minimum amount required to invest, there is no such minimum required to invest in LSVCCs. Third, individuals investing in LSVCCs receive tax benefits from both the federal and provincial governments. For example, the federal government provides a tax credit of 15%. The provincial governments also provide a tax credit although amounts may

differ. In Ontario, for example, the provincial government also provides an additional 15% provincial tax credit in addition to the federal tax credit. Fourth, all LSVCCs require at least one member of the board directors to be from a labour union, hence the labour sponsorship. In addition, the provincial and/or federal legislations often limit both the types of the companies and the amount of investment that the fund can invest in each company. Last, LSVCCs often have geographic restrictions on the location of both investee firms and investors. Due to its structure, Cumming and MacIntosh (2006) and Cumming and Johan (2009) argue that the resulting agency problems and costs have lead to poor investment returns.

With many commonalities among different LSVCCs, there are also many differences. First, many provincial LSVCCs are only allowed to accept investments from the local, provincial residents except for LSVCCs formed in Quebec or federally incorporated LSVCCs. Second, minimum holding periods or lock-up periods vary in different provinces. For example, LSVCC investors may only execute redemptions 8 years after initial investment. Manitoba legislation however requires a minimum lock-up of seven years and in Quebec, investors will not be able to redeem their shares until their retirement.

To support the development of LSVCCs in Canada, both federal and provincial governments provide generous income tax credits for individual investors. In Ontario for example, prior to 2007, individual investors are able to claim a 15% tax credit on a maximum investment up to Can\$5,000. After 2007, the maximum investment amount was increased to \$7,500. This means that individuals are able to claim up to \$750 in tax credits before 2007 and up to \$1,125 in tax credits from 2007 and beyond.¹ To further support investment in innovation, the Ontario government also provides an additional 5% tax credit if investment is made in research-oriented investment funds. The federal government provided a similar benefit in income

tax benefits. In other words, individual investors could enjoy up to an immediate 30% tax benefit (returns) after their investment, even before any investment profits are accounted for. In effect, depending on the individual's tax bracket, the real cost of a \$5,000 investment in LSVCCs through a Registered Retirement Savings Plan (a type of government registered savings plan which allows account holders to save for their retirement on a tax-deferred basis) may only range from \$1,180 to \$2,390 (Cumming and MacIntosh, 2006). The tax credits not only made the high risk investment in the venture capital funds more attractive, but arguably it negated the need for individual investors to be overly attentive over their investment (as RRSP investments are for retirement purposes, hence long term in horizon).

On September 2005, the Ontario government announced that they intended to remove the tax credit for investments in LSVCCs at the end of the 2010 tax year. In May 2008, the Ontario government delayed such process and planned to eliminate all the tax credits by the end of the 2011 tax year². The plan was to gradually cut the tax credit by 5% each year starting the tax year 2010 and to totally eliminate the tax credit by the tax year 2012. Also, from 2005, the LSVCC was only required to invest 60 percent of capital raised in private entrepreneurial firms or small-medium sized companies instead of the 70 percent as original required. In the next section we will discuss why we believe such reform will result in changes to LSVCC behaviour within Ontario.

LSVVC GOVERNANCE STRUCTURE AND FUND MANAGERS' INVESTMENT BEHAVIOUR

We believe that there are at least two theoretical reasons to explain how eliminating the tax credits could affect the fund managers' investment behaviour. First, individual investors are

incentivized to invest with a promise of tax credits due to the cheaper actual cost of initial investment and more importantly are less concerned about the long-term after tax return due to the up-front financial benefits (Cumming and MacIntosh, 2003 a, b; Cumming and Johan, 2009). While Barclay et al. (1998) argues that exiting fund investors prefer to defer the taxable capital gains, Bergstresser and Poterba (2002) find that after-tax returns affect investors' investment decisions and explain better net mutual fund inflow. Before the tax reform, LSVCCs in Ontario benefitted from the substantial tax credits (up to 30%) so much so that a negative portfolio (pre-tax) return could potentially turn into a substantial positive return after tax savings for its investors. Cumming and MacIntosh (2003a,b) show that investors lower their required rate return from LSVCCs investment when compared with both mutual funds or private equity fund investment. It follows therefore that, eliminating the tax credit will not only make LSVCCs *less* attractive for future investors, but existing investors will seek higher required rates of return for their investment. As such, the fund managers may change portfolio risk, measured by volatility, by drifting away from the less volatile investments in private entrepreneurial firms towards more volatile public companies, or the fund manager could continue to provide venture capital to risky firms with huge potential upside returns (positively skewed returns distribution) as their initial investment mandate required.

Hypothesis 1: After the announcement of the elimination the tax credit, LSVCCs in Ontario will style drift, while LSVCCs in other provinces will not.

Second, the special tax treatment given to LSVCCs has resulted in a competitive advantage over other forms of investment funds availed to retail investors such as mutual funds. With the

removal of the tax benefits, the pre- and after-tax returns expectations will converge. Given the relatively poor average performance of LSVCCs in Ontario, this will most likely translate into an underperformance, with expected withdrawals from investors as a result. LSVCC managers in Ontario anticipating this scenario have an incentive to change their investment behaviour and increase portfolio risk in order to improve expected performance (not necessarily risk-adjusted performance) and increase liquidity. Similar behavior by mutual fund managers is referred to as tournament behavior (see, for instance, Brown et al., 1996 and Chevalier and Ellison, 1997).

Similar tournament behaviour within LSVCCs in Ontario may be demonstrated as a result of various factors. Rather than using detailed, highly frequent and timely data, individual investors tend to pay more attention to annual performance rankings. As there are lock-ups in place, and most LSVCC investments are made within registered retirement portfolios to maximize tax benefits, individual investors may not necessarily pay attention to such rankings on an annual basis. Empirical evidence has also shown a convex relationship between fund flows and past performance for mutual funds (Sirri and Tufano, 1998; Del Guercio and Tkac, 2002). ‘Underperformers’ tend to be less affected by outflows than ‘outperformers’ are affected by fund inflows. As a result of tax benefits, long term investment horizons and lock-ups, such relationship between fund flows and performance may be similar for LSVCCs. LSVCCs are similar to mutual funds in that individual investors are allowed to invest, albeit with maximum investment restrictions (typically up to \$5000), and this results in a highly dispersed investor structure which apparently reduces the incentive to closely monitor fund managers due to coordination costs and free-rider problems (see Cumming and MacIntosh, 2006). The inclusion of the restrictive lock-ups of up to eight years also clearly reduces the incentive to monitor fund managers as investors are unable to redeem their investments in the event of unsatisfactory

performance. Also, the LSVCC structure is designed as open-ended instead of the more standard, limited life-time of about ten years typical of other venture capital funds. The ten year time limit essentially puts in place a schedule whereby fund managers must meet agreed upon performance measures related to investments, monitoring and mentoring of investees and exits by the end of the life of the fund. Fund managers that are unable to do so will face extreme difficulties raising capital for follow-up funds. Open-ended LSVCCs have no equivalent time frame, other than the lock-ups, to provide such discipline. Finally, each LSVCC fund is required to have a labour union sponsor or director. The labour sponsorship was implemented to ensure that the LSVCCs remain faithful to the initiative to create innovative provincial companies that would foster employment, not to necessarily provide value-added assistance to fund managers. Unfortunately such provincial-centric sponsorship by unions with no ownership interest in the funds themselves may have only served to hinder the LSVCCs in their investment activities, and more specifically provided no added corporate governance other than to ensure the maintenance of statutory restrictions (Cumming and MacIntosh, 2003c). Less frequent monitoring by retail investors, labour union board members and disproportionate fund flows can only impede effective fund manager governance and encourage a drift in investment behaviour. Recall that LSVCC fund managers are experts in investing in venture capital, and any drift away from this form of capital may not necessarily benefit fund investors. Therefore, if instead of continuing to provide venture capital to risky firms with huge potential upside returns (positively skewed returns distribution) as their initial investment mandate required, thus maximizing risk-adjusted returns, the LSVCC managers in Ontario could start drifting away from the less volatile investments in private entrepreneurial firms towards more volatile public companies taking on added risk (increase in return distribution's variance) in the hopes of improving their relative performance compared to

other LSVCCs or mutual funds as well as their competitors. Such move from venture capital to public equities could also be made to ensure sufficient liquidity in the event of potential redemptions. Regardless of motive, such tournament behaviour may not necessarily benefit existing investors although the rather restricted corporate governance as explained above may actually facilitate it. As such:

Hypothesis 2: After the announcement, LSVCCs in Ontario will increase their investments in public companies and increase their risk levels (measured by volatility), while LSVCCs in other provinces will not.

In addition, it has to be noted that only Ontario has carried out such tax reform with the removal of the tax credits. Across other Canadian provinces, LSVCCs were created with a similar common objective of providing venture capital to SME firms to facilitate economic development and growth, and tax credits to individual investors were also provided. Unlike the government of Ontario, other provinces continue with their support of their LSVCCs. This enables us to carry out a multi-jurisdictional comparison of behaviour among fund managers in Canada. Our third hypothesis is developed based on this discussion:

Hypothesis 3: LSVCCs in Ontario are expected to drift away from their original mandates, while LSVCCs in other province are expected to hold or increase their venture capital investment levels in private firms

DATA DESCRIPTIONS AND SUMMARY STATISTICS

The data used in this paper comprises 41 LSVCCs across Canada. Like many other types of investment funds, LSVCCs are required to publish audited annual reports to their shareholders. We reviewed all available LSVCC annual reports and hand collected the data based on information obtained from these reports. The data comprises information from 1997 to 2011 and the main dependent variables are the number of public companies in the fund portfolios at the end of the fiscal year versus the number of private firms in the fund portfolios at the end of the fiscal year. We also measure the percentage of investments, as the percent of overall investment, in public companies, at market value, percentage of investment in private firms at fair market value and percentage of short-term investment or marketable securities of fixed term investment with a term maturity of less than one year at market value. Cumming (2006) finds both fund and investee characteristics have an influence on the portfolio size of venture capital funds. We obtained several proxies for LSVCCs fund characteristics from their annual reports, including Fund Age which is the number of years since the LSVCC incorporated up to 2011, the Management Expense Ratio which is the total fees paid to fund managers and other parties as a percentage of the total assets of the fund, the Log (Total Assets) which is the log of the fund total assets in US\$ millions in the lagged period, and the Return on Assets (ROA) which is the net income over total assets at the end of the year. We also obtained the type and stage of investee firms from various sources. A dummy variable equal to 1 is used to indicate Early Stage firms, and the same dummy variable is used to indicate High-Technology firms. As additional control variables, Cumming and MacIntosh (2006) find that both bank interest rates and overall stock market performance have significant impact on the supply as well as the demand of the LSVCC

funds. We obtained the bank interest rate in the lagged period from the Bank of Canada. Similar to the proxy used in Cumming and MacIntosh (2006), we gather the SP/TSX return in the lagged period from Bloomberg and calculate the index annual returns. We also collect log of provincial GDP growth data from Statistics Canada.

The graphs in Figures 1 and 2 illustrate the LSVCC's investment behaviour over time. Prior to 2001, LSVCCs hardly invested in any public companies. After 2001, we see a gradual increase in the number investments in public companies in the portfolios. We especially observe a clear upward trend in terms of the number of public companies in the portfolios for LSVCCs in Ontario. There is also an increase in number of investments in public companies in the portfolios LSVCCs in other provinces but we observe a sharp decrease after 2010. From Figure 2, we find that LSVCCs in other provinces invest in more private companies than LSVCCs in Ontario.

Insert Figure 1 and Figure 2 about here

Table 2 provides descriptive statistics for the full sample of the data. The summary statistics show that LSVCC funds invest heavily in private firms both in terms of number and in the percentage of total amount invested. On average, LSVCCs invest in approximately 21 private companies and weight over 60% of total investment in the private firms. This is not surprising as the primary objective of LSVCCs is to provide venture capital to private entrepreneurial firms. In addition, the data shows that LSVCCs are rewarded higher

management fees. We find that the average management expense ratio is 3% (more than double that of mutual funds). This number is very similar to the 4% found in study by Cumming and MacIntosh (2006). Extant research (MacIntosh, 1997; Cumming and MacIntosh, 2006b; Osborne and Sandler, 1998) have found that LSVCCs provide lower returns relative to other type of investment vehicles. In our data, we find that on average, the return for LSVCCs in Canada is -4% per annum.

Insert Table 1 about here

UNIVARIATE TESTS

Table 3 provides a comparison of means and medians tests of the number of public companies; number of private firms in the portfolios; and Return on Asset (ROA) before and after the announcement of the change in tax policy on 2005, respectively.

Insert Table 2 about here

Panel A of Table 3 reports the difference in means and medians of number of public companies in the fund portfolio in general; in the Ontario LSVCC funds; and in the funds of all other

provinces, respectively. The mean (median) value of the number of public companies in the fund portfolios are 1.04 (0) before the announcement of tax policy changes in Ontario in 2005. (Before 2005, there after) and are 6.80 (1) after the announcement of tax policy change in Ontario (After 2005, there after). Our comparisons show the changes in the number of public companies in the fund portfolios in terms of both mean and median value are statistically significant at the 1% level. The picture is even clearer after we separate the funds by their location: LSVCCs funds that are in Ontario and LSVCCs funds that are in all other provinces. For funds in Ontario, the mean (median) number of public companies in the portfolios are 1.07 (0) before 2005 and 8.20 (2) after 2005. There is a drift from private companies and a clear increase of investment in public companies for the Ontario LSVCCs, which supports Hypothesis 1 and Hypothesis 2. Our statistical tests support this view. The statistical tests for difference in mean and median show that this is significant at 1% for both the mean and median numbers. For funds in other provinces, the mean (median) number of public companies in the portfolios are 1 (0) before 2005 and 3.44 (0) after 2005. The statistical tests show that although there is an increase of investment in the public companies for LSVCCs in other provinces, it is only significant in the mean but not significant in the median.

Table 3 Panel B shows the average (median) value is 20.48 (13) before 2005 and 20.81 (16) after 2005 for the number of private companies in the fund portfolios. Separating funds by their location, the mean (median) value is 15.49 (12.5) before 2005 for funds in Ontario, 30.75 (18) for funds in other provinces before 2005. After 2005, the mean (median) value is 18.19 (16) for funds in Ontario and 27.05 (15) for funds in the other provinces. From the statistics, we observe a slight increase in the number of private companies in the fund portfolios in Ontario and

a slight decrease in the number of private companies in the fund portfolios in other provinces. However, none of our statistical tests come up to be statistically significant.

Table 3 Panel C shows the average (median) value of Return on Assets (ROA) for the full sample of data. On average (median), the LSVCCs provide -2% (-1%) before 2005 and -5% (-2%) after 2005. Specifically, funds in Ontario on average (median) have an ROA of -1% (-1%) before 2005 and -6% (-2%) after 2005. The difference is at 10% significance for the average test. For funds in other provinces, the average (median) ROA is -2% (-1%) before 2005 and -3% (-1%) after 2005. However, we cannot find any statistically significant difference between pre-2005 and post-2005 for funds in other provinces. Nevertheless, we find that the risks (in terms of standard deviation) in returns are much greater for funds in Ontario after 2005, the same as our prediction in Hypothesis 2. On the contrary, we find that the risks in return decrease for funds in other provinces from before 2005 to after 2005 period.

In line with Hypothesis 1 and Hypothesis 2, the comparison results show that LSVCCs in Ontario have drifted from investing in private companies and invested in more public companies after the announcement of changes in tax policies and have undertaken more risk after the announcement as well. This type of univariate comparison is not fully informative since we have not yet controlled for other factors to be equal, particularly in regard to both economic and market conditions across different provinces. Our empirical tests in the next section include variables that have been found to have influences on fund investment and return.

Table 4 presents a correlation matrix for the main variables used in the multivariate tests. From the correlation matrix, we find a positive relationship between the investment in public companies and the change tax policy in Ontario. In addition, the correlation matrix also shows

that there is a statistically significant correlation structure among several variables. In this case, we present alternative specification with and without collinear variables in the regression.

Insert Table 3 about here

MULTIVARIATE ANALYSES

To understand the impact of the changes in the tax policy, with respect to investment strategy, we conduct empirical tests to show that there is a change in the components of the portfolios after 2005 (see Table 5). We test the same eight different models with and without controlling for investee characteristics, fund characteristics and macroeconomic variables for both Panels in Table 5. The dependent variable is the number of public companies in the portfolios for Panel A and the dependent variable is the number of private firms in the portfolios for Panel B. Model 1 and Model 2 run a simple difference-in-difference (DID) regression on the tax policy change in 2005 with different cluster controls. Model 3 and Model 4 show the results with control on investee characteristics and fund characteristics, respectively. Model 5 shows the results with macro-economic variables and Model 6 present the results with all control variables. For the last two models, Model 7 and Model 8, we repeat model 6 and test the results with funds in Ontario and funds in other provinces, respectively.

Insert Table 4 about here

The results in Table 5 Panel A consistently show that after the change in tax policy in 2005, LSVCCs funds are more likely to include public companies in their portfolios. The effect is significant at least at 10% level—supporting Hypothesis 1 and Hypothesis 2. In the most conservative estimate, after the tax policy change, there is an increase of 2.46 more public companies in the portfolios. The average number of public companies in the fund portfolios is 4.16. After the tax policy change, there is a 59.13% ($2.46/4.16=0.5913$) increase in the number of public companies in the fund portfolios. Since the main objective of the LSVCCs is to provide venture capital to private entrepreneurial firms, increasing investment in public companies crucially undermine such objective. From a diversification perspective, individual investors expected their investment in a venture capital fund in lieu of a mutual fund (a pooled investment in less liquid private companies rather than in the public companies invested in by mutual funds) to diversify their own personal portfolios. This style drift instead may result in investors unknowingly taking on higher (unwanted) systematic risk. LSVCCs may be becoming more mutual-like than initially planned and expected. As a robustness check, we separated the funds between funds in Ontario and funds in other provinces and retest the Model 6. The results show that the changes in the tax policies in Ontario have significant impact on LSVCCs funds in Ontario but have no such impact on the funds in other provinces. These results may not be surprising since changes in the tax policies in Ontario should only affect the individual investors in Ontario and not others. In terms of numbers, we find that LSVCCs in Ontario are more likely to double ($4.14/4.16=0.9952$) their number of public companies in the portfolios. We do not

observe such effects for funds in other provinces. These results correspond well to our second hypothesis that after the announcement of the elimination the tax credit, LSVCCs in Ontario will increase their investments in public companies, while LSVCCs in other province will not..

Other than the main variables of interest, our results also find that LSVCC funds that focus on early stage private firms are less likely to invest in public companies. In terms of provinces, Ontario LSVCCs with early stage focus are less likely to invest in public companies; however, LSVCCs in other provinces with early stage focus are more likely to invest in public companies. Both results are statistically significant at 5%. Similar to the early stage focus, Ontario LSVCCs with a focus on technology firms are less likely invest in the public firms but LSVCCs in other provinces with technology focus will be more likely to invest in public firms. We also find limited evidence that fund age, size and bank rate have an impact on the fund investment in the public companies but these results are not consistent.

Table 5 Panel B illustrates the regression results for the number of private companies in the portfolio. In general, we could not observe any effect on investment in the private companies after the tax policy change announcement in Ontario. When separating the funds by funds in Ontario and funds in other provinces, we find that LSVCCs in Ontario are more likely to reduce their number of private companies in their fund portfolios. On average, LSVCCs in Ontario have reduced the number of private companies in their portfolios by 2.72. The average number is 20.66 and this shows that after the tax policy change, there is a 13.17% ($2.72/20.66=0.1317$) decrease in investment in terms of the number of private companies. These results are statistically significant at the 1% level. LSVCCs in other provinces have reduced the number of private companies in their portfolios by 0.77, however, we are unable to find any statistical significance.

Besides the policy variable we are interested in, the results in Panel B Table 5 also show some other interesting findings. We find that LSVCCs with a focus on investment in the early stage firms are less likely to increase in their investment in the private companies. We find that high management expense ratios (MER) are positively associated with investment in private companies and larger funds are more likely to increase investment in terms of the number of private companies in the fund portfolio. Short term interest rates are positively associated with the number of private companies in the portfolios as well. We find limited evidence that as the funds grow older, the fund is more likely to invest in the private companies. However, these results are only statistically significant for funds in Ontario. A similar story applies to the size of the funds.

So far, we have only tested the level of public and private companies in the fund portfolios. As a robustness check, we also test whether there is a change in the fund investment allocation before and after the changes of tax policies in Ontario on 2005. The results are illustrated in Table 6. Table 6 provides the regression results on the percentage of total investment in the public/private companies as well as in short term investments. The dependent variables are percent of total investment in public companies, percentage of total investment in private companies, and percentage of total investment in short term. Standard errors are clustered by year for all models. Model 1 presents the results for LSVCC funds in Ontario only and Model 2 presents the results for LSVCC funds in all other provinces.

In terms of percent of total investment (measured in Canadian dollars) in public companies, LSVCCs in Ontario are more likely to place heavier weight in public companies and LSVCCs in other provinces are more likely to reduce the weight in public companies. Our results show that there is an increase of 6 percentage points as an investment increase in public

companies for LSVCCs in Ontario. Considering the average percentage of total investment in the public companies is 12%, this means a 50% ($0.06/0.12=0.5$) increase in the weight of investment in public companies for LSVCCs in Ontario. There is also a reduction of 7 percentage points for investments in public companies for LSVCCs in other provinces. The average amount of public companies is 12% and this represents a 58.33% ($0.07/0.12=0.5833$) reduction the weight of the investment in public companies for the LSVCCs in other province. Both results are statistically significant at the 10% level. Other than the main variable of interests, we also find that LSVCCs in Ontario with a focus on early stage and technology are less likely to invest in the public companies and LSVCCs in other provinces with a focus on early stage and technology are more likely to invest in public companies. Bank rates are statistically significantly negatively related to the investment in public companies for Ontario LSVCCs only. Fund age, stock market return and local economic development are positively related to the percentage of total investment in the public companies only for the funds that are outside the province of Ontario.

Insert Table 5 about here

We find mixed results in terms of the percentage investment in private companies and in terms of the percentage investment in short-term/government bonds (see Table 6). In general, after the announcement of the tax changes in Ontario, we find that LSVCCs in other provinces are more likely to invest in private companies but found no such differences for LSVCCs in

Ontario. Ontario LSVCCs seem to shift investment in the short-term/government bond to the public companies however we do not observe such shifts for LSVCCs in other provinces. So far, our empirical results have indicated that only LSVCCs in Ontario are shifting away from their original mandates to invest in entrepreneurial firms and to promote local economic growth, measured either by the number of and the percentage of private firms in their portfolios which we interpret as clear support for Hypothesis 3. Besides our main variable of interest, we also find that the fund size, as measured in terms of total assets, has a positive impact levels of investment in private companies for both LSVCCs in Ontario and LSVCCs in other provinces. Similarly, past fund returns as measured in terms of ROA has a positive impact on the short term investment for LSVCCs in provinces outside Ontario. Also, LSVCCs in other provinces are less likely to put money in the short-term /government bonds as markets improve. However, we could not find a similar effect for LSVCCs in Ontario.

ROBUSTNESS CHECK

On May 14, 2008, the Ontario government announced a delay of the phase-out of the tax credit by additional year. In this section, we test whether the trend for Ontario LSVCCs to increasingly invest in public companies became even stronger after the announcement in 2008. Table 7 provides an additional robustness check in relation to the trend for Ontario LSVCCs to invest in public companies after the second announcement in 2008. The dependent variables are number of public and private companies in the portfolios, percentage of total investment in public/private companies and short term investments. Standard errors are clustered by year for

all models. Model 1 present the results for LSVCC funds in Ontario only and Model 2 presents the results for LSVCC funds in all other provinces.

Insert Table 6 about here

Our results in Table show that there is an increase of 8.15 more public companies in the fund portfolios. In term of percentage of total investment in the public companies, we find that there is a 10.5 percentage point increase. On the other hand, LSVCCs in other provinces were more likely to increase the percentage of total investment in private companies by 23.3. All those results are both economically and statistically significant. The tendency for fund managers in Ontario to invest in public companies strengthened even more with the imminent the phase-out of the tax credits, which is also in line with our prediction our three hypotheses.

CONCLUSION

The primary purpose of this paper is to understand the dual roles played by government policy and the resulting inefficient corporate governance structure of LSVCCs in the portfolio choices of LSVCCs in Canada, and especially the ones situated in the province of Ontario. We know that government support has been found to be crucial for bridging the capital gap faced by nascent entrepreneurial firms and for the provision of venture capital (Di Giacomo, 2004; Duanjie, et al., 2002; Laliberte, 1998; Cumming and Johan, 2009). However, without proper

mechanisms and efficient governance structures, government sponsored programs may not be able to achieve such objectives. In this case, the generous tax subsidies initially provided to incentivise retail investors to invest in LSVCCs may have been misdirected as they resulted in disincentives for the same retail investors to direct the appropriate corporate governance measures against fund managers. The tax credits essentially mitigated the need for disparate investors to monitor fund managers as up-front benefits had already been obtained, and any need for returns postponed as such investments were mainly made with long-term benefits sought within retirement accounts. Even as LSVCCs have been found to be underperforming even risk free 30 day Treasury Bills (Cumming and MacIntosh, 2006, 2007; Cumming and Johan, 2009), the usual disciplining method used against other financial intermediaries could not be used as redemptions have been prohibited as a result of lock-ups. As a result of lack of oversight or attention, the announcement of tax reform could only bring about changes that may not necessarily be beneficial for investors. We show that the announcement of the impending change in tax policy in Ontario (2005) have resulted in a style drift by the Ontario government sponsored LSVCCs. LSVCCs in Ontario are more likely to include more public companies in the fund portfolios and reduce their holdings of private firms after the 2005, which not only reduces the diversification benefits of (retail) investors, but also increases portfolio risk as well as other LSVCC risks as fund managers are experts in venture investment, not investment in public equities. Such drift however may be necessary to ensure increased liquidity as redemptions may increase pursuant to the lock-up periods expiring. On the other hand, we find no evidence on the changes of weighting of public and private companies within the fund portfolio for LSVCCs in other provinces. Instead, the LSVCCs in other province are decreasing the percentage of investments in public companies and increasing the percentage of total investment to private

companies within a similar period. We argue that two theoretical reasons could explain this tendency of fund managers in Ontario to invest more in public companies. LSVCC managers in Ontario may initiate tournament behaviour, drift away from their mandate to invest in venture firms and invest in more risky publicly listed companies (measured by volatility), to counter or mitigate the potential effect of the change in investor sentiment pursuant to the removal of the tax credit, and the potential redemptions as the lock-up periods expire. Such risky style drift may actually be facilitated by the lack of oversight by retail investors as they are not incentivized to monitor their investments and the performance of fund managers. Albeit paying LSVCC fund managers fees close to double those of mutual fund managers, investors' diversification benefits are further eroded by the style drift. Increasing investment in publicly listed companies result in an even wider funding gap for entrepreneurs seeking venture capital funding. Extant research have found that retail-based venture capital funds set up with the best of intentions have failed to accomplish what they were established for. We add to the literature with our findings relating to the Canadian experiment and hope that our findings will encourage future research to help create programs that will accomplish their goals.

ENDNOTES

¹ However, unused tax credits cannot be carried over to the future .See <http://www.fin.gov.on.ca/en/guides/lsif.html>.

² See <http://www.fin.gov.on.ca/en/bulletins/lsif/0206.html>.

REFERENCES

- Barclay, M. J., Pearson, N. D. & Weisbach, M. S. 1998. Open-end mutual funds and capital-gains taxes, *Journal of Financial Economics*, 49: 3-43.
- Bergstresser, D. & Poterba, J. 2002. Do after-tax returns affect mutual fund inflows? *Journal of Financial Economics*, 63: 381-414.
- Brown, K. C., Harlow, W. V. & Starks, L. T. 1996. Of tournaments and temptations: An analysis of managerial incentives in the mutual fund industry, *Journal of Finance*, 51: 85-110.
- Chevalier, J. & Ellison, G. 1997. Risk taking by mutual funds as a response to incentives, *Journal of Political Economy*, 105: 1167-1200.
- Cumming, D. 2006a. Adverse selection and capital structure: Evidence from Venture Capital, *Entrepreneurship: Theory & Practice*, 30: 155-183.
- Cumming, D. 2006b. The determinants of venture capital portfolio size: Empirical evidence, *Journal of Business*, 79: 1083-1126.
- Cumming, D. 2007. Government policy towards entrepreneurial finance: Innovation investment funds, *Journal of Business Venturing*, 22: 193-235.
- Cumming, D. 2011. Public policy and the creation of active venture capital markets, *Venture Capital*, 13: 75-94.
- Cumming, D., Fleming, G. & Schwenbacher, A. 2008. Financial intermediaries, ownership structure and the provision of venture capital to SMEs: evidence from Japan, *Small Business Economics*, 31: 59-92.
- Cumming, D., Fleming, G. & Suchard, J. A. 2005. Venture capitalist value-added activities, fundraising and drawdowns, *Journal of Banking & Finance*, 29: 295-331.
- Cumming, D., Hass, L. H. & Schweizer, D. 2013. Private Equity benchmarks and portfolio optimization, Available at SSRN: <http://ssrn.com/abstract=1687380>.
- Cumming, D. J. & Johan, S.A. 2012. **Venture Capital and Private Equity Contracting: An International Perspective (Second Edition)**. London: Elsevier Science Academic Press.
- Cumming, D. J. & Johan, S.A. 2010. Phasing Out an Inefficient Venture Capital Tax Credit, *Journal of Industry, Competition & Trade*. 10: 227-252.
- Cumming, D. J. & Johan, S. A. 2012. Is Venture Capital in Crisis? *World Economic Review*. July/August: 69-72.

- Cumming, D., Schmidt, D. & Walz, U. 2010. Legality and venture capital governance around the world, *Journal of Business Venturing*, 25: 54-72.
- Cumming, D. J. & MacIntosh, J. G. 2003a. A cross-country comparison of full and partial venture capital exits, *Journal of Banking & Finance*, 27: 511-548.
- Cumming, D. & MacIntosh, J. G. 2003b. Venture Capital Exits in Canada and the United States, *University of Toronto Law Journal*, 53: 101-200.
- Cumming, D.J., & J. MacIntosh, 2003c. Canadian Labor-Sponsored Venture Capital Corporations: Bane or Boon? In A. Ginsberg and I. Hasan (Eds.), **New Venture Investment: Choices and Consequences**: 169-200, Elsevier Science Academic Press, London.
- Cumming, D. J. & MacIntosh, J. G. 2005. Comparative venture capital governance: Private versus labour sponsored venture capital funds, *Venture Capital, Entrepreneurship and Public Policy*. 9: 69-94.
- Cumming, D. J. & MacIntosh, J. G. 2006. Crowding out private equity: Canadian evidence, *Journal of Business Venturing*. 21: 569-609.
- Cumming, D. J. & MacIntosh, J. G. 2007. Mutual funds that invest in private equity? An analysis of labour-sponsored investment funds, *Cambridge Journal of Economics*. 31: 445-487.
- Del Guercio, D. & Tkac, P. A. 2002. The determinants of the flow of funds of managed portfolios: Mutual funds vs. pension funds. *Journal of Financial and Quantitative Analysis*. 37: 523-557.
- Di Giacomo, M. 2004. Public support to entrepreneurial firms: An assessment of the role of Venture Capital in the European experience, *Journal of Private Equity*, 8: 22-38.
- Duanjie, C., Lee, F. C. & Mintz, J. 2002. Taxation, SMEs and Entrepreneurship, *OECD Papers*.
- Florou, A. 2005. Discussion of Performance of Private to Public MBOs: The Role of Venture Capital, *Journal of Business Finance & Accounting*, 32: 683-690.
- Laliberte, P. 1998. The economic impact of labour-sponsored venture capital corporations: A case study from Quebec, *International Labour Review*, 137: 47-60.
- OECD, R. 2004. Venture Capital: Trends and policy recommendations, *OECD Papers*, 4: 1-29.
- Sandler, D. 2004. **Venture Capital and Tax Incentives: A Comparative Study of Canada and the United States**. Canadian Tax Foundation.

Sirri, E. R. & Tufano, P. 1998. Costly Search and Mutual Fund Flows, *Journal of Finance*, 53: 1589-1622.

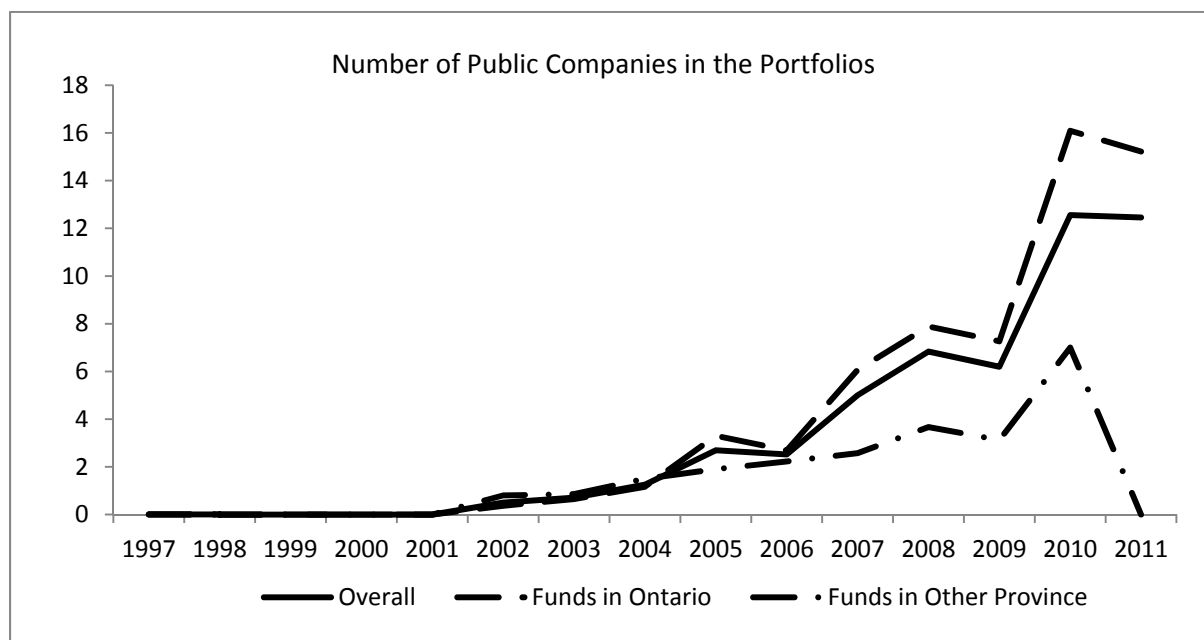


Figure 1: This figure presents the average number of public companies in the fund portfolio between 1997 and 2011.

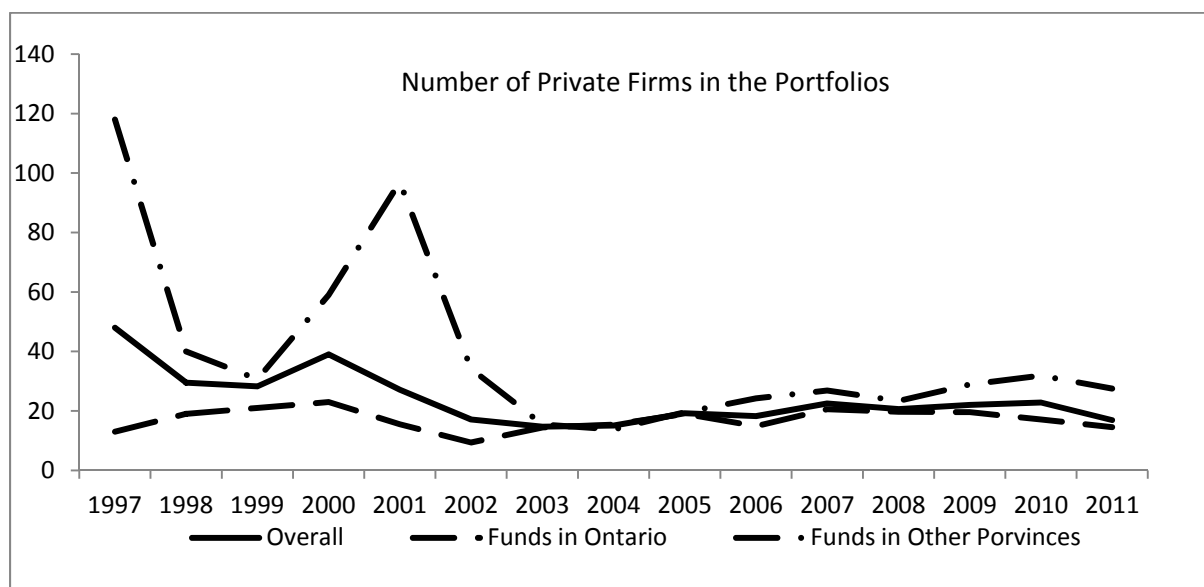


Figure 2: This figure presents the average number of private firms in the fund portfolio between 1997 and 2011.

TABLE 1 DESCRIPTIVE STATISTICS

This table present statistics for the full sample of fund-year observations in the data. The data range from 1997 to 2011.

Variable Name	Mean	Median	Standard Deviation	Minimum	Maximum	Number of Observations
# Public Companies in the Portfolios	4.16	0.00	11.54	0.00	83.00	244
# Private Firms in the Portfolios	20.66	15.00	22.18	0.00	132.00	242
% Investment in Public Companies	0.12	0.00	0.25	0.00	1.00	244
% Investment in Private Firms	0.60	0.61	0.24	0.00	1.00	244
% Investment in Short-term	0.28	0.28	0.23	0.00	1.00	244
Treat	0.68	1.00	0.47	0.00	1.00	244
After 2005	0.54	1.00	0.50	0.00	1.00	244
Treat * After 2005	0.38	0.00	0.49	0.00	1.00	244
Early Stage	0.25	0.00	0.44	0.00	1.00	244
Technology	0.41	0.00	0.49	0.00	1.00	244
Fund Age	6.01	6.00	4.02	0.00	18.00	244
Management Expense Ratio	0.03	0.02	0.14	0.00	2.24	241
Log (Total Assets)	17.56	17.65	1.45	10.52	20.44	215
Return On Assets (ROA)	-0.04	-0.01	0.17	-1.15	0.35	244
Bank Rate	3.27	3.19	1.26	0.65	5.77	244
SP/TSX Return	0.06	0.12	0.21	-0.35	0.31	244
Log (Provincial GDP Growth)	0.02	0.02	0.02	-0.03	0.08	238

TABLE 2 COMPARISON TESTS

This table presents the comparison of mean and median test for the number of public companies in the portfolios (Panel A), number of private companies in the portfolios (Panel B) and Return On Assets (Panel C). The *, ** and *** denote the significance at 10%, 5% and 1% level, respectively.

Panel A: Number of Public Companies in the Portfolios

	All Funds		Ontario Funds		Other Province Funds	
	Before 2005	After 2005	Before 2005	After 2005	Before 2005	After 2005
Group	0	1	0	1	0	1
Number of Observations	112	132	74	93	38	39
Mean	1.04	6.80	1.07	8.20	1.00	3.44
Median	0	1	0	2	0	0
Standard Deviation	3.80	14.81	4.44	16.87	2.08	7.11
Difference in Mean (0-1)	-4.298***		-3.913***		-2.050**	
Difference in Median (0-1)	p=0.000***		p=0.000***		p=0.112	

Panel B: Number of Private Firms in the Portfolios

	All Funds		Ontario Funds		Other Province Funds	
	Before 2005	After 2005	Before 2005	After 2005	Before 2005	After 2005
Group	0	1	0	1	0	1
Number of Observations	110	132	74	93	36	39
Mean	20.48	20.81	15.49	18.19	30.75	27.05
Median	13	16	12.5	16	18	15
Standard Deviation	22.49	22.00	13.45	16.32	32.21	31.09
Difference in Mean (0-1)	-0.114		-1.175		0.505	
Difference in Median (0-1)	p=0.411		p=0.239		p=0.213	

Panel C: Return On Assets (ROA)

	All Funds		Ontario Funds		Other Province Funds	
	Before 2005	After 2005	Before 2005	After 2005	Before 2005	After 2005
Group	0	1	0	1	0	1
Number of Observations	112	132	74	93	38	39
Mean	-0.02	-0.05	-0.01	-0.06	-0.02	-0.03
Median	-0.01	-0.02	-0.01	-0.02	-0.01	-0.01
Standard Deviation	0.12	0.20	0.11	0.23	0.13	0.10
Difference in Mean (0-1)	1.796*		1.915*		0.144	
Difference in Median (0-1)	p=0.304		p=0.188		p=0.910	

TABLE 3 CORRELATION MATRIX

This table presents Pearson correlation coefficients for the full sample of fund-year in the data. The * indicates the correlations are statistically significant at 5%.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) # Public Companies in the Portfolios	1																
(2) # Private Firms in the Portfolios	-0.105	1															
(3) % Investment in Public Companies	0.808*	-0.231*	1														
(4) % Investment in Private Firms	-0.477*	0.223*	-0.545*	1													
(5) % Investment in Short-term	-0.388*	0.0231	-0.525*	-0.428*	1												
(6) Treat	0.111	-0.265*	0.0142	0.068	0.084 3	1											
(7) After 2005	0.241*	-0.0342	0.243*	0.183*	0.448 *	0.075 1	1										
(8) Treat * After 2005	0.254*	-0.154*	0.190*	0.134	0.341 *	0.562 *	0.733 *	1									
(9) Early Stage	-0.140*	-0.208*	-0.0779	0.0017 3	0.082 2	0.042 2	-0.117	0.097 2	1								
(10) Technology	-0.173*	-0.0275	-0.112	0.153*	- 0.035	-0.133	- 0.036 9	- 0.150 *	0.043 4	1							
(11) Fund Age	0.126	0.212*	0.125	0.211*	- 0.349 *	0.021 9	0.457 *	0.325 *	- 0.246 *	0.007 3	1						
(12) Management Expense Ratio	-0.00611	-0.0569	-0.0139	-0.0688	0.084 8	0.028 8	- 0.086 4	- 0.074 4	- 0.055 6	- 0.056 7	-0.097	1					
(13) Log (Total Assets)	-0.132	0.613*	-0.186*	0.251*	- 0.054 5	- 0.040 7	0.009 4	- 0.000 3	- 0.076 5	0.036 6	0.268 *	- 0.362 *	1				
(14) Return On Assets (ROA)	0.039	0.01	0.0672	-0.114	0.043 2	0.060 2	0.094 7	-0.131	-0.112	- 0.186 *	0.063 6	0.010 2	0.035 1	1			
(15) Bank Rate	-0.183*	0.129	-0.196*	-0.0368	0.248 *	-0.032	-0.114	- 0.073 4	0.069 8	0.063 9	- 0.248 *	-0.041	0.203 *	- 0.223 *	1		
(16) SP/TSX Return	0.0229	-0.0061	0.0406	-0.0598	0.017	0.064 2	0.045 9	0.081 8	0.049 2	0.094 1	-0.117	0.051 5	0.017 3	0.238 *	- 0.166*	1	
(17) Log (Provincial GDP Growth)	-0.193*	0.0236	-0.139*	-0.0932	0.244 *	0.157 *	0.389 *	0.393 *	0.029 9	0.030 4	- 0.286 *	0.036 6	0.054	0.121	0.340*	0.27 6*	1

TABLE 5 REGRESSION RESULTS

This table presents Ordinary Least Square (OLS) panel regressions of determinates of the number of public companies (Panel A) and number private companies (Panel B) in the portfolio. Variables are defined as in Table 1. Standard errors are clustered by year for Model 1, 5, 6, 7, and 8 and are clustered by province and year for Model 2, 3, and 4. Model 1 and model 2 present the difference-in-difference (DID) regression results. Model 3, 4, 5 present the results with investee characteristics, fund characteristics and Macro/Economic Variables, respectively. Model 6 presents the results with all variables. Model 7 presents the results with funds in Ontario only and model 8 presents the results with funds in all other provinces. The *, ** and *** are statistically significant at the 10%, 5% and 1% level, respectively. T-statistics are in square brackets.

Panel A: Number of Public Companies in the Portfolio								
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
	DID	DID	Investee Characteristics	Fund Characteristics	Macro/Economic Variables	All Variables	Ontario Fund Only	Other Provincial Funds
Constant	1.00*** [3.33]	1.03* [1.71]	2.95 [1.65]	25.139*** [3.14]	6.14*** [4.17]	28.13** [2.35]	33.68* [2.01]	-7.26 [-0.44]
Treat	0.07 [0.30]	0.04 [0.11]	0.41 [0.30]	0.050 [0.12]	0.01 [0.02]	0.38 [0.83]		
After 2005	2.44** [2.94]	2.91** [2.08]	3.34* [1.97]	2.07 [1.03]	2.00** [2.43]	2.60** [2.24]	4.14* [1.82]	-1.23 [-0.61]
Treat*After2005	4.70*** [3.82]	4.22*** [14.48]	3.00** [2.31]	4.25** [2.36]	4.08*** [2.98]	2.46* [1.88]		
Investee Characteristics								
Early Stage			-2.75** [-2.25]			-3.34*** [-3.79]	-4.83** [-3.71]	2.35** [2.83]
Technology			-2.90 [-1.04]			-3.30** [-2.33]	-6.69** [-3.39]	6.15** [2.81]
Fund Characteristics								
Fund Age				0.25** [2.46]		0.04 [0.24]	0.095 [0.59]	0.91* [2.00]
Management Expense Ratio				-3.18 [-1.47]		-4.30 [-1.21]	-6.09 [-1.56]	-3.88 [-0.06]
Log (Total Assets)				-1.42*** [-3.19]		-1.17 [-1.72]	-1.36 [-1.44]	0.11 [0.15]
Return On Assets (ROA)				4.68** [2.44]		0.19 [0.06]	0.05 [0.01]	-2.66 [-0.87]
Macro/Economic Variables								
Bank Rate					-1.28** [-2.70]	-0.95* [-1.88]	-0.92 [-1.14]	-0.32 [-0.93]
SP/TSX Return					2.16 [1.04]	0.96 [0.46]	-0.44 [-0.18]	3.54 [1.39]
Log(Provincial GDP Growth)					-24.89 [-0.78]	-32.88 [-0.87]	-33.84 [-0.65]	-11.10 [-0.43]
Observations	244	238	238	210	241	210	148	62
R-Squared	0.08	0.08	0.11	0.10	0.08	0.14	0.18	0.35
Cluster Control	Year	Province and Year	Province and Year	Province and Year	Year	Year	Year	Year

Panel B: Number of Private Firms in the Portfolio

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
	DID	DID	Investee Characteristics	Fund Characteristics	Macro/ Economic Variables	All Variables	Ontario Fund Only	Other Provincial Funds
Constant	30.75*** [4.57]	31.54* [2.40]	34.96* [2.51]	-145.83** [-2.55]	24.67*** [4.73]	-139.93*** [-7.25]	-87.53*** [-6.00]	-327.97*** [-6.97]
Treat	-15.26** [-2.23]	-16.06 [-1.22]	-14.45 [-1.15]	-11.94 [-1.57]	-15.98** [-2.56]	-10.78*** [-3.07]		
After 2005	-3.70 [-0.54]	-2.75 [-0.23]	-2.31 [-0.19]	-2.11 [-0.38]	-2.82 [-0.45]	-1.51 [-0.38]	-2.72*** [-3.04]	-0.77 [-0.17]
Treat*After2005	6.41 [0.92]	5.46 [0.46]	2.82 [0.23]	0.38 [0.06]	4.35 [0.66]	-2.88 [-0.69]		
Investee Characteristics								
Early Stage			-10.66** [-2.70]			-6.86*** [-3.37]	-5.38*** [-4.23]	-7.71*** [-4.067]
Technology			-2.29 [-0.39]			-3.36 [-1.48]	0.43 [0.17]	-5.98 [-1.01]
Fund Characteristics								
Fung Age				0.47 [1.69]		0.32* [1.88]	0.54*** [3.66]	-0.89 [-0.69]
Management Expense Ratio				27.03** [2.59]		24.64*** [6.15]	13.10*** [4.86]	734.38** [2.54]
Log (Total Assets)				9.85*** [3.05]		9.67*** [8.09]	5.79*** [6.40]	19.94*** [8.96]
Return On Assets (ROA)				2.20 [0.34]		-0.23 [-0.04]	6.59*** [3.43]	11.40 [0.60]
Macro/Economic Variables								
Bank Rate					2.69** [2.72]	0.92* [1.76]	1.08** [2.24]	2.01** [2.43]
SP/TSX Return					10.60 [1.70]	2.12 [0.73]	-2.34 [-1.66]	4.17 [0.80]
Log(Provincial GDP Growth)					-116.43 [-1.74]	-87.56* [-1.78]	-8.88 [-0.39]	-396.47*** [-3.82]
Observations	242	236	236	208	236	208	148	60
R-Squared	0.07	0.08	0.12	0.47	0.10	0.50	0.39	0.73
Cluster Control	Year	Province and Year	Province and Year	Province and Year	Year	Year	Year	Year

TABLE 5 REGRESSION RESULTS ON THE PERCENTAGE OF TOTAL INVESTMENT IN PUBLIC/PRIVATE COMPANIES AS WELL AS IN SHORT TERM INVESTMENTS

This table presents Ordinary Least Square (OLS) panel regression of Percentage of Total Investment in Public/Private companies and in short term investment. Variables are defined as in Table 1. Standard errors are clustered by years for all models. Model 1 presents the result for LSVCC funds in Ontario only and model 2 presents the results for LSVCC funds in other provinces. The *, ** and *** are statistically significant at the 10%, 5% and 1% level, respectively. T- statistics are in square brackets.

	Percent of Total Investment in Public Companies		Percent of Total Investment in Private Firms		Percent of Total Investment in Short Term	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
	Ontario Fund Only	Other Provincial Funds	Ontario Fund Only	Other Provincial Funds	Ontario Fund Only	Other Provincial Funds
Constant	0.47* [2.04]	1.04*** [3.57]	-0.00 [-0.00]	013 [0.45]	0.53* [1.70]	-0.16 [-0.32]
After 2005	0.06* [1.95]	-0.07* [-1.94]	0.07 [1.53]	0.23** [2.57]	-0.13** [-2.51]	-0.16 [-1.05]
Investee Characteristics						
Early Stage	-0.09*** [-4.74]	0.16*** [5.28]	0.04 [0.62]	-0.06 [-1.23]	0.05 [0.57]	-0.10 [-0.95]
Technology	-0.14*** [-4.45]	0.24*** [4.69]	0.12** [2.63]	-0.10 [-1.72]	0.02 [0.57]	-0.14 [-1.08]
Fund Characteristics						
Fung Age	0.004 [1.12]	0.04** [2.71]	0.004 [0.53]	-0.03* [-2.16]	-0.01 [-1.17]	-0.006 [-0.27]
Management Expense Ratio	-0.10 [-1.73]	-3.63 [-1.26]	0.07 [1.47]	-1.81 [-0.35]	0.03 [0.38]	5.45 [1.35]
Log (Total Assets)	-0.01 [-1.11]	-0.07*** [-4.01]	0.03** [2.12]	0.05** [3.06]	-0.02 [-0.94]	0.02 [0.72]
Return On Assets (ROA)	0.01 [0.11]	-0.22 [-1.67]	-0.07 [-0.88]	-0.06 [-0.41]	0.06 [0.72]	0.28*** [3.53]
Macro/Economic Variables						
Bank Rate	-0.03*** [-3.20]	0.004 [0.26]	-0.02 [-0.89]	-0.04* [-1.81]	0.04*** [3.01]	0.03 [1.05]
SP/TSX Return	-0.02 [-0.40]	0.14** [2.26]	0.004 [0.12]	-0.22* [-1.87]	0.01 [0.20]	0.08 [1.16]
Log(Provincial GDP Growth)	-0.42 [-0.71]	2.77** [3.12]	0.75 [1.43]	-1.60 [-0.84]	-0.33 [-0.74]	-1.17* [-1.97]
Observations	148	62	148	62	148	62
R-Squared	0.24	0.49	0.17	0.40	0.34	0.31
Cluster Control	Year	Year	Year	Year	Year	Year

TABLE 6 ROBUSTNESS CHECKS

This table presents Ordinary Least Square (OLS) panel regression of Number and Percentage of Total Investment in Public/Private companies and in short term investment. Variables are defined as in Table 1. Standard errors are clustered by years for all models. Model 1 presents the result for LSVCC funds in Ontario only and model 2 presents the results for LSVCC funds in other provinces. The *, ** and *** are statistically significant at the 10%, 5% and 1% level, respectively. T- statistics are in square brackets.

	Number of Public Companies		Number of Public Companies		Percent of Total Investment in Public Companies		Percent of Total Investment in Private Firms		Percent of Total Investment in Short Term	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
	Ontario Fund Only	Other Provincial Funds	Ontario Fund Only	Other Provincial Funds	Ontario Fund Only	Other Provincial Funds	Ontario Fund Only	Other Provincial Funds	Ontario Fund Only	Other Provincial Funds
Constant	29.76 [1.77]	-7.101 [-0.44]	-86.88*** [-6.13]	-334.3*** [-6.62]	0.428* [1.79]	1.069*** [3.50]	0.0153 [0.05]	-0.0159 [-0.07]	0.556*** [3.37]	-0.0529 [-0.12]
After 2008	8.148*** [3.62]	0.167 [0.08]	-3.310 [-1.38]	10.42 [1.34]	0.105*** [3.15]	-0.0462 [-1.13]	0.0517 [0.91]	0.233*** [3.77]	-0.156** [-2.53]	-0.187*** [-4.45]
Investee Characteristics										
Early Stage	-4.660*** [-3.33]	2.016** [2.91]	-5.370*** [-4.19]	-8.385*** [-3.40]	-0.0862*** [-4.39]	0.148*** [4.51]	0.0334 [1.65]	-0.0182 [-0.45]	0.0527** [2.33]	-0.130** [-2.44]
Technology	-6.941*** [-3.55]	5.820** [2.82]	0.672 [0.26]	-7.755 [-1.30]	-0.148*** [-4.90]	0.224*** [4.43]	0.112*** [3.05]	-0.0738 [-1.34]	0.0359* [1.83]	-0.150*** [-3.84]
Fund Characteristics										
Fung Age	0.0903 [0.91]	0.727** [2.54]	0.496*** [3.39]	-1.427 [-1.69]	0.00431 [1.44]	0.0286** [2.18]	0.00628* [1.84]	-0.00714 [-0.59]	-0.0106*** [-3.47]	-0.0215** [-2.24]
Management Expense Ratio	-5.822 [-1.48]	-1.829 [-0.03]	13.25*** [5.06]	805.9** [2.54]	-0.0939 [-1.68]	-3.832 [-1.28]	0.0557 [0.70]	-0.688 [-0.17]	0.0382 [0.97]	4.521 [0.75]
Log (Total Assets)	-1.437 [-1.47]	0.150 [0.18]	5.860*** [6.38]	19.86*** [8.80]	-0.0159 [-1.18]	-0.0689*** [-3.80]	0.0292* [1.83]	0.0354** [2.72]	-0.0133 [-1.67]	0.0334 [1.66]
Return On Assets (ROA)	-1.026 [-0.23]	-2.933 [-1.08]	7.210*** [3.05]	11.90 [0.70]	-0.00403 [-0.04]	-0.233 [-1.77]	-0.0852 [-1.43]	-0.00761 [-0.03]	0.0892 [1.45]	0.240 [1.16]
Macro/Economic Variables										
Bank Rate	0.636 [0.77]	-0.377 [-0.82]	0.363 [0.54]	3.847** [2.18]	-0.00441 [-0.43]	-0.00834 [-0.45]	-0.00341 [-0.23]	0.0207 [0.76]	0.00782 [0.41]	-0.0124 [-0.61]
SP/TSX Return	6.654** [2.71]	3.221* [1.94]	-5.557* [-1.80]	10.55 [1.32]	0.0777* [2.11]	0.0852 [1.48]	0.0651 [0.85]	0.0159 [0.17]	-0.143* [-2.00]	-0.101 [-1.62]
Log(Provincial GDP Growth)	-14.64 [-0.36]	-7.677 [-0.25]	-7.688 [-0.22]	-343.9** [-2.73]	-0.243 [-0.47]	2.667*** [3.08]	0.489 [0.60]	-0.824 [-0.77]	-0.246 [-0.25]	-1.842 [-1.16]
Observations	148	62	148	60	148	62	148	62	148	62
R-Squared	0.183	0.346	0.386	0.731	0.238	0.486	0.156	0.317	0.307	0.287
Cluster Control	Year	Year	Year	Year	Year	Year	Year	Year	Year	Year